Autonomous Ocean Drones for Coral Monitoring

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Coral reef degradation remains a significant challenge for marine ecosystems worldwide. Traditional monitoring approaches often require extensive fieldwork, specialized equipment, and substantial travel costs. The CoralEye initiative explored whether autonomous systems could provide broader coverage while maintaining scientific accuracy and operational efficiency.

Each drone used a hybrid solar-wave power system capable of operating continuously for up to 11 months. The drones captured high-resolution imagery, environmental measurements, and reef health indicators. Data was transmitted to centralized processing systems where automated classification models evaluated reef conditions and highlighted anomalies requiring further investigation.

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